

CM

Installation and operating instructions



Installation and operating instructions

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GRUNDFOS 

English (GB) Installation and operating instructions

Original installation and operating instructions

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Warning
Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.



Warning
The use of this product requires experience with and knowledge of the product.
Persons with reduced physical, sensory or mental capabilities must not use this product, unless they are under supervision or have been instructed in the use of the product by a person responsible for their safety.
Children must not use or play with this product.

1. Symbols used in this document



Warning
If these safety instructions are not observed, it may result in personal injury.



Warning
If these instructions are not observed, it may lead to electric shock with consequent risk of serious personal injury or death.



Warning
The surface of the product may be so hot that it may cause burns or personal injury.



Caution
If these safety instructions are not observed, it may result in malfunction or damage to the equipment.



Note
Notes or instructions that make the job easier and ensure safe operation.

2. Introduction

This manual describes the installation and operation of Grundfos CM pumps.

3. Delivery and handling

The pumps are delivered from factory in a packaging specially designed for manual transport or transport by forklift truck or a similar vehicle.

Note

In order to ensure safe transport, we recommend that you transport the pumps with suitable lifting tools.

4. Applications

The pumps are horizontal, multistage centrifugal pumps designed for pumping of clean, thin and non-flammable liquids, not containing solid particles or fibres that may attack the pump mechanically or chemically.



Warning

The pump must not be used for the transfer of flammable or toxic liquids.

5. Identification

5.1 Nameplates for pump

The pump name plates are positioned on the motor fan cover or terminal box.

5.1.1 Nameplate with pump data

The data and information on the pump nameplate are described in the table below. See the nameplate in fig. 1 on page 14.

Pos.	Description
1	Pump type
2	Pump model
3	Maximum ambient temperature
4	Temperature class
5	Minimum efficiency index
6	Maximum system pressure
7	Maximum liquid temperature
8	Hydraulic efficiency at best efficiency point
9	Insulation class
10	Motor protection
11	Rated flow
12	Head at rated flow
13	Maximum head

5.1.2 Nameplate with approval marks

The data and information on the pump nameplate are described in the table below. See the nameplate in fig. 2 on page 14.

Pos.	Description
1	CE mark
2	EAC mark
3	PSE mark
4	cULus mark
5	UL mark
6	cURus mark
7	Company name and address
8	Country of manufacture

5.2 Nameplate for motor

The motor name plate is positioned on the motor cooling fins.

The data and information on the motor nameplate are described in the table below. See the nameplate in fig. 3 on page 14.

Pos.	Description
1	Capacitor size and voltage
2	50 Hz motor efficiency at rated work point
3	50 Hz power factor
4	50 Hz output power in kW
5	Frequency
6	Number of phases
7	50 Hz output power in hp
8	50 Hz maximum current
9	50 Hz full-load current
10	50 Hz rated voltage
11	Motor type
12	50 Hz rated speed
13	Frequency
14	60 Hz output power in kW
15	NEMA enclosure class
16	60 Hz output power in hp
17	60 Hz power factor
18	60 Hz motor efficiency at rated work point
19	Part number
20	Factory code
21	Production date (year and week)
22	Country of origin
23	60 Hz rated voltage
24	60 Hz full-load current
25	60 Hz maximum current
26	60 Hz rated speed
27	IEC duty cycle
28	Number of poles
29	IEC enclosure class
30	Insulation class
31	NEMA enclosure type
32	Motor duty class
33	Maximum ambient temperature
34	NEMA locked-rotor code
35	NEMA design class
37	CC122B mark
38	CE mark
39	cURus mark

6. Mechanical installation

Before installing the pump, check that the pump type and parts are as ordered.



Warning

When pumping hot or cold liquids, make sure that persons cannot accidentally come into contact with hot or cold surfaces.

6.1 Installation of pump

Install the pump on a plane surface using the mounting holes in the motor base plate and a minimum of four bolts. Tighten each of the four bolts to a torque of 10 Nm.

Install the pump so that air locks are avoided in the pump housing and pipework.

Figure 1 and the table below show the permissible pump positions.

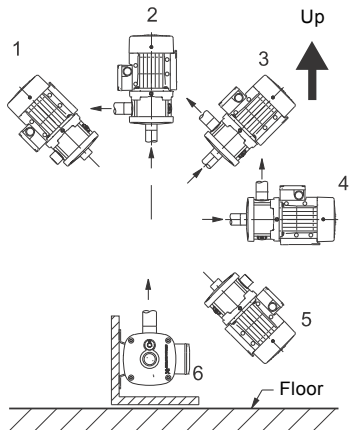


Fig. 1 Pump positions

Pump position	Non-self-priming pumps	Self-priming pumps
1	-	-
2	•	-
3	•	-
4	•	•
5	-	-
6	•	•

- Mounting in this position is allowed.
- Install the pump so that inspection, maintenance and service can easily be performed.
- Install the pump in a well-ventilated location.

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6.2 Pipework

We recommend that you fit isolating valves on either side of the pump. It is thus not necessary to drain the system if the pump needs service.

If the pump is installed above the liquid level, a non-return valve must be fitted in the suction pipe below the liquid level. See fig. 4.

Self-priming pumps:

Note

We recommend an opening pressure of the non-return valve which is lower than 0.05 bar. Otherwise the additional resistance will reduce the suction capability of the pump.

If the pump is to be used for pumping rainwater or well water, we recommend that you fit a filter to the inlet of the suction pipe.

The pump must not be stressed by the pipework.

Install the pipes according to the design requirements given in EN ISO 13480-3:2012. Tolerances must comply with EN ISO 13920:1996, class C.

The pipework must be correctly sized taking due account of the pump inlet pressure.

Install the pipes so that air locks are avoided, especially on the suction side of the pump. See fig. 2.

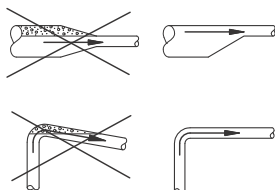


Fig. 2 Pipework

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6.2.1 Pipe connection (non-self-priming pumps)

Take care not to damage the pump when connecting the suction and discharge pipes.

Caution

Torque: 50-60 Nm. The stated torque must not be exceeded.

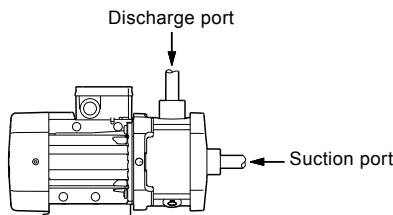


Fig. 3 Suction and discharge ports

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6.2.2 Pipe connection (self-priming pumps)

The pump must be installed correctly to ensure that it can self-prime.

Take the following precautions:

See fig. 4.

- The minimum height from the centre of the suction port to the first tapping point (H_1) must be observed. If a pressure manager is installed in the system, H_1 is the height from the centre of the pump suction port to the pressure manager. Minimum heights appear from the table below.
- The suction pipe must be at least 0.5 metres below the liquid level (H_3).

For optimum suction capability, the pump must be located near the well or tank to ensure that the suction pipe is as short as possible. This will reduce the self-priming time, especially in the case of a high suction lift.

Note

We recommend that you install a filling plug in the discharge pipe. This facilitates liquid filling before startup. See fig. 4, pos. A.

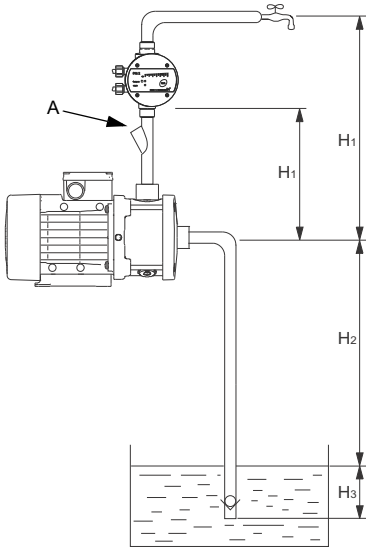


Fig. 4 Recommended pipework for a self-priming pump

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Suction lift (H_2) [m]	Minimum height (H_1) [m]
4	0.2
5	0.35
6	0.5
7	0.6
8	0.7

6.3 Alternative connection positions

The pump is available with various connection positions on special request. See fig. 5.

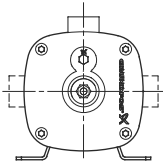


Fig. 5 Alternative connection positions

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Self-priming pumps:

Note These pumps are only available with the discharge port pointing upwards, i.e. in the same direction as the filling hole.

6.4 Terminal box positions

The pump is available with various terminal box positions on special request. See fig. 6.

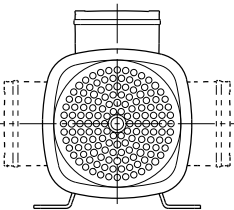


Fig. 6 Terminal box positions

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6.5 Avoiding condensation in the motor

If the liquid temperature falls below the ambient temperature, condensation may form in the motor during standstill. Condensation can occur in moist surroundings or areas with high humidity.

In such cases, use a motor suitable for condensing environments such as an IPX5 motor available from Grundfos.

Alternatively, open the bottom drain hole in the motor flange by removing the plug. See fig. 7. This reduces the motor enclosure class to IPX5.

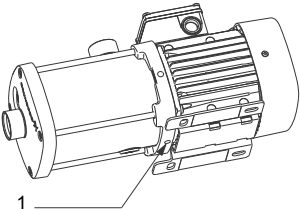


Fig. 7 Motor drain plug

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Pos.	Description
1	Motor drain plug

The open drain hole helps prevent condensation in the motor as it makes the motor self-venting and allows water and humid air to escape.

When you install the pump outdoors, provide the motor with a cover to avoid condensation. See fig. 8.

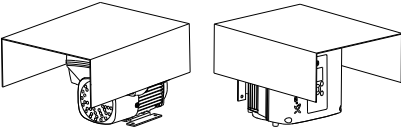


Fig. 8 Examples of covers (not supplied by Grundfos)

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7. Electrical installation

Carry out the electrical connection according to local regulations.

Check that the supply voltage and frequency correspond to the values stated on the nameplate.

Warning

The electrical connection must be carried out in accordance with local regulations.

Before starting work on the pump, switch off the power supply. Make sure that the power supply cannot be accidentally switched on.



The pump must be connected to an external all-pole mains switch according to local regulations.

The product must be earthed and protected against indirect contact in accordance with local regulations.

Wires connected to supply terminals, must be separated from each other and from the supply by reinforced insulation.

7.1 Power supply cable

In order to comply with the EN 60335-1 standard, the power supply cable must as minimum be rated for an operating temperature of 105 °C (221 °F).

7.2 Motor protection

Single-phase motors, 1 x 115 / 230 V, 60 Hz

These motors do not incorporate motor protection and must be connected to a motor-protective circuit breaker which can be manually reset.

Set the motor-protective circuit breaker to maximum $1.15 \times I_{1/1}$.

Other single-phase motors

These motors have built-in current- and temperature-dependent motor protection in accordance with IEC 60034-11 and require no further motor protection. The motor protection is of the TP 211 type which reacts to both slow- and quick-rising temperatures. The motor protection is automatically reset.

Three-phase motors up to 3 kW

These motors must be connected to a motor-protective circuit breaker which can be manually reset.

Set the motor-protective circuit breaker to maximum 1.15 times full-load current.

Three-phase motors of 3 kW and up

These motors have built-in thermistors (PTC)*. The thermistors are designed according to DIN 44082. The motor protection is of the TP 211 type which reacts to both slow- and quick-rising temperatures.

* Applies only to motors for the following supply voltages:

- 3 x 200 V / 346 V, 50 Hz
- 3 x 200-220 V / 346-380 V, 60 Hz
- 3 x 220-240 V / 380-415 V, 50 Hz.

Motors for other supply voltages must be connected to a motor-protective circuit breaker as described for three-phase motors up to 3 kW.

7.3 Electrical connection

Carry out the electrical connection as shown in the diagram inside the terminal box cover.

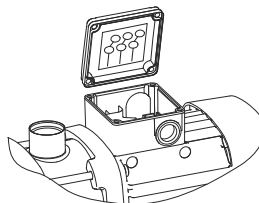


Fig. 9 Wiring diagram

7.4 Frequency converter operation

You can connect three-phase motors to a frequency converter.

Depending on the frequency converter type, this may cause increased acoustic noise from the motor. Furthermore, it may cause the motor to be exposed to detrimental voltage peaks.

MG 71- and MG 80-based motors have no phase insulation* and must therefore be protected against voltage peaks higher than 650 V (peak value) between the supply terminals.

Caution

* MG 71- and MG 80-based motors with phase insulation are available on request.

The above disturbances, i.e. both increased acoustic noise and detrimental voltage peaks, can be eliminated by fitting an LC filter between the frequency converter and the motor.

For further information, please contact the frequency converter supplier or Grundfos.

Self-priming pumps only:

If the pump is connected to a frequency converter, operation at low speed may cause the internal recirculation valve to open. This will result in a drop in pressure and flow.

Note

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8. Startup

Note

If there is a risk of condensation in the motor, remove the motor drain plug before startup and keep the drain hole open during operation. See fig. 7.

8.1 Non-self-priming pumps

Caution

Do not start the pump until it has been filled with liquid.

8.1.1 Liquid filling

Warning



Pay attention to the direction of the vent hole, and make sure that the escaping hot or cold liquid does not cause injury to persons or damage to the equipment.

1. Close the isolating valve on the discharge side of the pump.
2. Open the isolating valve in the suction pipe completely before starting the pump.
3. Remove the filling plug. See fig. 10.
4. Fill the pump housing and the suction pipe completely with liquid until a steady stream of liquid runs out of the filling hole.
5. Fit and tighten the filling plug.
6. Start the pump and slowly open the discharge isolating valve while the pump is running. This ensures venting and pressure build-up during startup.

The discharge isolating valve must be opened immediately after startup of the pump. Otherwise the temperature of the pumped liquid may become too high and cause damage to the equipment.

Caution

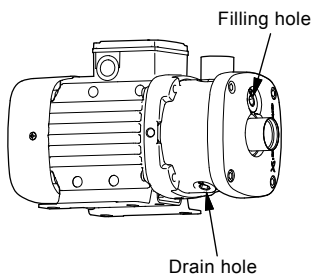


Fig. 10 Position of filling hole and drain hole

Note

If it is difficult for the pump to build up pressure, it may be necessary to repeat steps 1 to 6.

8.2 Self-priming pumps

Caution

Do not start the pump until it has been filled with liquid.

8.2.1 Liquid filling

Warning



Pay attention to the direction of the vent hole, and make sure that the escaping hot or cold liquid does not cause injury to persons or damage to the equipment.

1. Make sure that the discharge pipe is empty and that the height from the centre of the suction port to the first tapping point (H_1) meets the requirements. See section 6.2.2 *Pipe connection (self-priming pumps)*.
2. Open the isolating valves in the suction and discharge pipes.
3. Open a tap close to the pump so that air can escape.
4. Remove the filling plug in the pump. See fig. 11.
5. If a filling plug has been installed in the discharge pipe, remove this plug and use this hole for filling. Otherwise use the filling hole in the pump.
6. Fill the pump housing and the suction pipe completely with liquid until a steady stream of liquid runs out of the filling hole.
7. Fit and tighten the filling plug(s).
8. Start the pump and wait until liquid is pumped. If you have used the filling hole in the pump, it may be necessary to repeat steps 1 to 8 to ensure that the pump is completely filled with liquid.

Note

If connected to a frequency converter, the pump must run at maximum speed (3450 min^{-1}) during startup.

9. If the pump does not operate properly after several start attempts, see section 12. *Fault finding*.

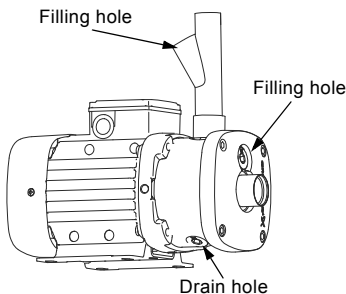


Fig. 11 Position of filling holes and drain hole

Note

The pump is allowed to run for 5 minutes to attempt to suck liquid. If the pump does not build up pressure and flow, repeat steps 1 to 8.

8.3 Checking the direction of rotation

Note The description below applies to three-phase motors only.

The motor fan cover has an installation indicator. See fig. 12. Based on the motor cooling air, it indicates the direction of rotation of the motor.

Before the motor is started for the first time or if the position of the indicator has been changed, the indicator function should be checked, for instance by moving the indicator field with a finger.

To determine whether the direction of rotation is correct or wrong, compare the indication with the table below.

Indicator field	Direction of rotation
Black	Correct
White/reflecting	Wrong*

* To reverse the direction of rotation, switch off the power supply and interchange any two of the incoming supply wires.

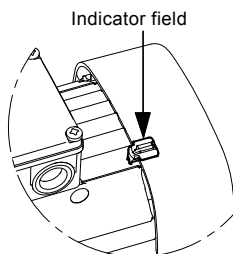


Fig. 12 Installation indicator

You can place the indicator in various positions on the motor, but do not place it between the cooling fins close to the screws that hold the fan cover.

The correct direction of rotation is also shown by arrows on the motor fan cover.

9. Maintenance

Warning



Before starting work on the pump, switch off the power supply. Make sure that the power supply cannot be accidentally switched on.

Warning



Make sure that the escaping water does not cause injury to persons or damage to the equipment.

The internal pump parts are maintenance-free. You must keep the motor clean in order to ensure adequate cooling of the motor. If the pump is installed in dusty environments, clean and check the pump regularly. Take the enclosure class of the motor into account when cleaning.

The motor has maintenance-free, greased-for-life bearings.

9.1 Frost protection

Pumps which are not being used during periods of frost must be drained to avoid damage.

Remove the filling and drain plugs from the pump. See fig. 10.

Do not refit the plugs until the pump is taken into operation again.

Caution

Before startup after a period of inactivity, the pump and the suction pipe must be completely filled with liquid. See section 8. *Startup*.

9.2 Cleaning

Prior to a long period of inactivity, flush the pump with clean water to prevent corrosion and deposits in the pump.

Use acetic acid to remove possible lime deposits from the pump.

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10. Service

Caution If used for a liquid which is injurious to health or toxic, the pump will be classified as contaminated.

Before the pump is returned to Grundfos for service, the safety declaration at the end of these instructions must be filled in by authorised personnel and attached to the pump in a visible position.

If Grundfos is requested to service the pump, it must be cleaned before it is returned.

If proper cleaning is not possible, all relevant information about the pumped liquid must be provided.

If the above is not fulfilled, Grundfos can refuse to accept the pump for service.

Possible costs of returning the pump are to be paid by the customer.

The safety declaration can be found at the end of these instructions (only in English).

11. Technical data

11.1 Enclosure class

- IP55 (standard)
- IPx5 (with motor drain plug removed)

11.2 Sound pressure level

The sound pressure level of the pumps is lower than 70 dB(A).

11.3 Ambient temperature

Self-priming pumps:
Caution The liquid temperature must not exceed 60 °C (140 °F).

Maximum ambient temperature	Liquid temperature
55 °C (131 °F) ²⁾	90 °C (194 °F) ^{1) + 2)}
50 °C (122 °F) ²⁾	100 °C (212 °F) ^{1) + 2)}
45 °C (113 °F)	110 °C (230 °F) ¹⁾
40 °C (104 °F)	120 °C (248 °F) ¹⁾

- 1) Only the stainless-steel variant (EN 1.4301 / AISI 304) is suitable for pumping liquids with temperatures above 90 °C (194 °F).
- 2) Does not apply for pumps with PSE approval (pumps approved for use in Japan).

If the ambient temperature exceeds 55 °C (45 °C for pumps with PSE approval), do not fully load the motor due to the risk of overheating. In such cases, you may need to derate the motor output or use an oversize motor with a higher rated output. You can derate the CM pumps in relation to the ambient temperature without any consequence. Contact Grundfos for further information. See fig. 13.

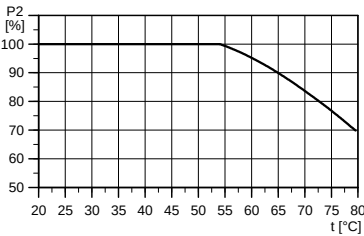


Fig. 13 Derating in relation to the ambient temperature

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11.4 Maximum system pressure and permissible liquid temperature

Material variant	Shaft seal	Permissible liquid temperature*		Maximum system pressure	
Cast iron (EN-GJL-200)	AVBx	-20 to 40 °C 41 to 90 °C	(-4 to 104 °F) (105.8 to 194 °F)	10 bar 6 bar	(145 psi) (87 psi)
	AQQx	-20 to 90 °C	(-4 to 194 °F)	10 bar	(145 psi)
Stainless steel (EN 1.4301 / AISI 304)	AVBx	-20 to 40 °C 41 to 90 °C	(-4 to 104 °F) (105.8 to 194 °F)	10 bar 6 bar	(145 psi) (87 psi)
	AQQx	-20*** to 90 °C 91 to 120 °C**	(-4 to 194 °F) (195.8 to 248 °F)	16 bar 10 bar	(232 psi) (145 psi)
Stainless steel (EN 1.4401 / AISI 316)	AVBx	-20 to 40 °C 41 to 90 °C	(-4 to 104 °F) (105.8 to 194 °F)	10 bar 6 bar	(145 psi) (87 psi)
	AQQx	-20*** to 90 °C 91 to 120 °C**	(-4 to 194 °F) (195.8 to 248 °F)	16 bar 10 bar	(232 psi) (145 psi)

* At liquid temperatures below 0 °C (32 °F), higher motor outputs may be needed due to increased viscosity, for instance if you have added glycol to the water.

** 120 °C applies only if the pump has an AQQE shaft seal.

*** CM pumps for pumping liquids at temperatures below -20 °C are available on request. Please contact Grundfos.

11.5 Minimum inlet pressure

You can calculate the minimum inlet pressure "H" in metres head required during operation to avoid cavitation in the pump from the following formula:

$$H = p_b \times 10.2 - \text{NPSH} - H_f - H_v - H_s$$

p_b = Barometric pressure in bar.
The barometric pressure can be set to 1 bar.
In closed systems, p_b indicates the system pressure in bar.

NPSH = Net Positive Suction Head in metres head. To be read from the NPSH curves on pages 15 to 17 at the highest flow the pump will be delivering.

H_f = Friction loss in suction pipe in metres head.

H_v = Vapour pressure in metres head.
See fig. 9, page 18.
 t_m = liquid temperature.

H_s = Safety margin = min. 0.5 metres head.

If the calculated value of "H" is positive, the pump can operate with a maximum suction lift of "H" metres.

If the calculated value of "H" is negative, a minimum suction head of "H" metres is required during operation to avoid cavitation.

Example

$p_b = 1$ bar.
Pump type: CM 3, 50 Hz.
Flow rate: 4 m³/h.
NPSH (from fig. 5, page 15): 3.3 metres head.
 $H_f = 3.0$ metres head.
Liquid temperature: 90 °C.
 H_v (from fig. 9, page 18): 7.2 metres head.
 $H = p_b \times 10.2 - \text{NPSH} - H_f - H_v - H_s$ [metres head].
 $H = 1 \times 10.2 - 3.0 - 3.3 - 7.2 - 0.5 = -3.8$ metres head.
This means that a suction head of 3.8 metres is required during operation.
Pressure calculated in bar: $3.8 \times 0.0981 = 0.37$ bar.
Pressure calculated in kPa: $3.8 \times 9.81 = 37.3$ kPa.

11.6 Maximum inlet pressure

The actual inlet pressure plus the pressure when the pump is operating against a closed valve must always be lower than the maximum system pressure.

12. Fault finding



Warning

Before removing the terminal box cover, switch off the power supply. Make sure that the power supply cannot be accidentally switched on.



Warning

The pumped liquid may be scalding hot and under high pressure. Before any removal or dismantling of the pump, the system must therefore be drained, or the isolating valves on either side of the pump must be closed.

Fault	Cause	Remedy
1. The pump does not run.	a) Supply failure.	Switch on the switch. Check cables and cable connections for defects and loose connections.
	b) Motor protection tripped.	See 2. a), b), c), d), e).
	c) Control-current circuit defective.	Repair or replace the control-current circuit.
2. Motor-protective circuit breaker has tripped (trips immediately when power supply is switched on).	a) Contacts of the motor-protective circuit breaker or magnet coil defective.	Replace the contacts of the motor-protective circuit breaker, the magnet coil or the entire motor-protective circuit breaker.
	b) Cable connection is loose or faulty.	Check cables and cable connections for defects, and replace the fuses.
	c) Motor winding is defective.	Repair or replace the motor.
	d) The pump is mechanically blocked.	Switch off the power supply, and clean or repair the pump.
	e) The setting of the motor-protective circuit breaker is too low.	Set the motor-protective circuit breaker according to the rated current of the motor ($I_{1/1}$). See nameplate.
3. The motor-protective circuit breaker trips occasionally.	a) The setting of the motor-protective circuit breaker is too low.	See 2. e).
	b) Periodic supply fault.	See 2. b).
	c) Periodically low voltage.	Check cables and cable connections for defects and loose connections. Check that the power supply cable of the pump is correctly sized.
4. The motor-protective circuit breaker has not tripped, but the pump is inadvertently out of operation.	a) See 1. a), b), c) and 2. d).	
5. The pump performance is unstable.	a) Pump inlet pressure too low.	Check the inlet conditions of the pump.
	b) Suction pipe is partly blocked by impurities.	Remove and clean the suction pipe.
	c) Leakage in suction pipe.	Remove and repair the suction pipe.
	d) Air in suction pipe or pump.	Vent the suction pipe or pump. Check the inlet conditions of the pump.
6. The pump performance is unstable, and the pump is noisy.	Self-priming pumps only:	
	a) The differential pressure across the pump is too low.	Close the tap gradually until the discharge pressure is stable and the noise has ceased.

Fault	Cause	Remedy
7. The pump runs, but gives no water.	a) Pump inlet pressure too low.	See 5. a).
	b) The suction pipe is partly clogged by impurities.	See 5. b).
	c) The foot or non-return valve is stuck in its closed position.	Remove and clean, repair or replace the valve.
	d) Leakage in suction pipe.	See 5. c).
	e) Air in suction pipe or pump.	See 5. d).
8. When startup is attempted, the pump will start, but delivers no pressure or flow.	Self-priming pumps only:	
	a) Liquid column above non-return valve in discharge pipe prevents the pump from self-priming.	Empty the discharge pipe. Make sure that the non-return valve does not hold back liquid in the discharge pipe. Repeat the startup procedure in section 6.2.2 Pipe connection (self-priming pumps) .
	b) Suction pipe draws in air.	Make sure that the suction pipe is airtight from pump to liquid level. Repeat the startup procedure in section 6.2.2 Pipe connection (self-priming pumps) .
9. The pump runs, but does not deliver the rated flow.	Self-priming pumps only:	
	a) The internal valve did not close.	Close the tap gradually until a sudden rise in pressure or flow can be seen. Then open the tap gradually until the required flow is reached.
10. The pump runs backwards when switched off.	a) Leakage in suction pipe.	See 5. c).
	b) Foot or non-return valve defective.	See 7. c).
	c) The foot valve is stuck in completely or partly open position.	See 7. c).
11. The pump runs with reduced performance.	a) Wrong direction of rotation.	Three-phase pumps only: Switch off the power supply with the external circuit breaker, and interchange two phases in the pump terminal box. See also section 8.3 Checking the direction of rotation .
	b) See 5. a), b), c), d).	

13. Further product information

13.1 Service documentation

Service documentation is available in Grundfos Product Center (<http://product-selection.grundfos.com/>).

If you have any questions, please contact the nearest Grundfos company or service workshop.

14. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

Appendix

Type	①		Tliq,max	⑦ °C ⑦ °F					
Model	②		PMax	⑥ bar ⑥ PSI ⑥ MPa					
TAmb	③ °C ③ °F	TF	④	MEI≥	⑤	η _p (%)	⑧	Insulation class	⑨ ⑩
50 Hz	Q nom	⑪ m ³ /h ⑪ GPM	60 Hz	Q nom	⑪ m ³ /h ⑪ GPM				
	H nom	⑫ m ⑫ PSI		H nom	⑫ m ⑫ PSI				
	H max	⑬ m ⑬ PSI		H max	⑬ m ⑬ PSI				

Fig. 1 Pump nameplate with data

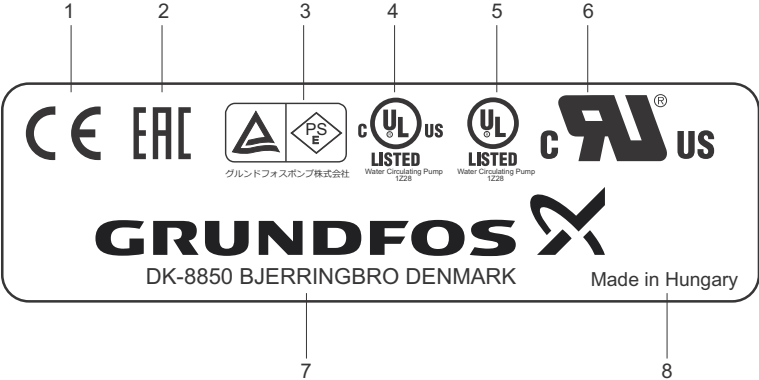


Fig. 2 Pump nameplate with approval marks

98811138	⑥ - MOT	Type	①	Env	⑮	Model	⑰ - ⑱ - ⑲	Country of origin IEC 60034 ⑳ ⑳ ㉑ ㉒ ㉓ ㉔ ㉕ ㉖ ㉗ ㉘ ㉙ ㉚ ㉛ ㉜ ㉝ ㉞ ㉟ ㊱ ㊲ ㊳ ㊴ ㊵ ㊶ ㊷ ㊸ ㊹ ㊺ ㊻ ㊼ ㊽ ㊾ ㊿ ㋀ ㋁ ㋂ ㋃ ㋄ ㋅ ㋆ ㋇ ㋈ ㋉ ㋊ ㋋ ㋌ ㋍ ㋎ ㋏ ㋐ ㋑ ㋒ ㋓ ㋔ ㋕ ㋖ ㋗ ㋘ ㋙ ㋚ ㋛ ㋜ ㋝ ㋞ ㋟ ㋠ ㋡ ㋢ ㋣ ㋤ ㋥ ㋦ ㋧ ㋨ ㋩ ㋪ ㋫ ㋬ ㋭ ㋮ ㋯ ㋰ ㋱ ㋲ ㋳ ㋴ ㋵ ㋶ ㋷ ㋸ ㋹ ㋺ ㋻ ㋼ ㋽ ㋾ ㋿ ㌀ ㌁ ㌂ ㌃ ㌄ ㌅ ㌆ ㌇ ㌈ ㌉ ㌊ ㌋ ㌌ ㌍ ㌎ ㌏ ㌐ ㌑ ㌒ ㌓ ㌔ ㌕ ㌖ ㌗ ㌘ ㌙ ㌚ ㌛ ㌜ ㌝ ㌞ ㌟ ㌠ ㌡ ㌢ ㌣ ㌤ ㌥ ㌦ ㌧ ㌨ ㌩ ㌪ ㌫ ㌬ ㌭ ㌮ ㌯ ㌰ ㌱ ㌲ ㌳ ㌴ ㌵ ㌶ ㌷ ㌸ ㌹ ㌺ ㌻ ㌼ ㌽ ㌾ ㌿ ㍀ ㍁ ㍂ ㍃ ㍄ ㍅ ㍆ ㍇ ㍈ ㍉ ㍊ ㍋ ㍌ ㍍ ㍎ ㍏ ㍐ ㍑ ㍒ ㍓ ㍔ ㍕ ㍖ ㍗ ㍘ ㍙ ㍚ ㍛ ㍜ ㍝ ㍞ ㍟ ㍠ ㍡ ㍢ ㍣ ㍤ ㍥ ㍦ ㍧ ㍨ ㍩ ㍪ ㍫ ㍬ ㍭ ㍮ ㍯ ㍰ ㍱ ㍲ ㍳ ㍴ ㍵ ㍶ ㍷ ㍸ ㍹ ㍺ ㍻ ㍼ ㍽ ㍾ ㍿ ㏀ ㏁ ㏂ ㏃ ㏄ ㏅ ㏆ ㏇ ㏈ ㏉ ㏊ ㏋ ㏌ ㏍ ㏎ ㏏ ㏐ ㏑ ㏒ ㏓ ㏔ ㏕ ㏖ ㏗ ㏘ ㏙ ㏚ ㏛ ㏜ ㏝ ㏞ ㏟ ㏠ ㏡ ㏢ ㏣ ㏤ ㏥ ㏦ ㏧ ㏨ ㏩ ㏪ ㏫ ㏬ ㏭ ㏮ ㏯ ㏰ ㏱ ㏲ ㏳ ㏴ ㏵ ㏶ ㏷ ㏸ ㏹ ㏺ ㏻ ㏼ ㏽ ㏾ ㏿ 㐀 㐁 㐂 㐃 㐄 㐅 㐆 㐇 㐈 㐉 㐊 㐋 㐌 㐍 㐎 㐏 㐐 㐑 㐒 㐓 㐔 㐕 㐖 㐗 㐘 㐙 㐚 㐛 㐜 㐝 㐞 㐟 㐠 㐡 㐢 㐣 㐤 㐥 㐦 㐧 㐨 㐩 㐪 㐫 㐬 㐭 㐮 㐯 㐰 㐱 㐲 㐳 㐴 㐵 㐶 㐷 㐸 㐹 㐺 㐻 㐼 㐽 㐾 㐿 㑀 㑁 㑂 㑃 㑄 㑅 㑆 㑇 㑈 㑉 㑊 㑋 㑌 㑍 㑎 㑏 㑐 㑑 㑒 㑓 㑔 㑕 㑖 㑗 㑘 㑙 㑚 㑛 㑜 㑝 㑞 㑟 㑠 㑡 㑢 㑣 㑤 㑥 㑦 㑧 㑨 㑩 㑪 㑫 㑬 㑭 㑮 㑯 㑰 㑱 㑲 㑳 㑴 㑵 㑶 㑷 㑸 㑹 㑺 㑻 㑼 㑽 㑾 㑿 㒀 㒁 㒂 㒃 㒄 㒅 㒆 㒇 㒈 㒉 㒊 㒋 㒌 㒍 㒎 㒏 㒐 㒑 㒒 㒓 㒔 㒕 㒖 㒗 㒘 㒙 㒚 㒛 㒜 㒝 㒞 㒟 㒠 㒡 㒢 㒣 㒤 㒥 㒦 㒧 㒨 㒩 㒪 㒫 㒬 㒭 㒮 㒯 㒰 㒱 㒲 㒳 㒴 㒵 㒶 㒷 㒸 㒹 㒺 㒻 㒼 㒽 㒾 㒿 㓀 㓁 㓂 㓃 㓄 㓅 㓆 㓇 㓈 㓉 㓊 㓋 㓌 㓍 㓎 㓏 㓐 㓑 㓒 㓓 㓔 㓕 㓖 㓗 㓘 㓙 㓚 㓛 㓜 㓝 㓞 㓟 㓠 㓡 㓢 㓣 㓤 㓥 㓦 㓧 㓨 㓩 㓪 㓫 㓬 㓭 㓮 㓯 㓰 㓱 㓲 㓳 㓴 㓵 㓶 㓷 㓸 㓹 㓺 㓻 㓼 㓽 㓾 㓿 㔀 㔁 㔂 㔃 㔄 㔅 㔆 㔇 㔈 㔉 㔊 㔋 㔌 㔍 㔎 㔏 㔐 㔑 㔒 㔓 㔔 㔕 㔖 㔗 㔘 㔙 㔚 㔛 㔜 㔝 㔞 㔟 㔠 㔡 㔢 㔣 㔤 㔥 㔦 㔧 㔨 㔩 㔪 㔫 㔬 㔭 㔮 㔯 㔰 㔱 㔲 㔳 㔴 㔵 㔶 㔷 㔸 㔹 㔺 㔻 㔼 㔽 㔾 㔿 㕀 㕁 㕂 㕃 㕄 㕅 㕆 㕇 㕈 㕉 㕊 㕋 㕌 㕍 㕎 㕏 㕐 㕑 㕒 㕓 㕔 㕕 㕖 㕗 㕘 㕙 㕚 㕛 㕜 㕝 㕞 㕟 㕠 㕡 㕢 㕣 㕤 㕥 㕦 㕧 㕨 㕩 㕪 㕫 㕬 㕭 㕮 㕯 㕰 㕱 㕲 㕳 㕴 㕵 㕶 㕷 㕸 㕹 㕺 㕻 㕼 㕽 㕾 㕿 㖀 㖁 㖂 㖃 㖄 㖅 㖆 㖇 㖈 㖉 㖊 㖋 㖌 㖍 㖎 㖏 㖐 㖑 㖒 㖓 㖔 㖕 㖖 㖗 㖘 㖙 㖚 㖛 㖜 㖝 㖞 㖟 㖠 㖡 㖢 㖣 㖤 㖥 㖦 㖧 㖨 㖩 㖪 㖫 㖬 㖭 㖮 㖯 㖰 㖱 㖲 㖳 㖴 㖵 㖶 㖷 㖸 㖹 㖺 㖻 㖼 㖽 㖾 㖿 㗀 㗁 㗂 㗃 㗄 㗅 㗆 㗇 㗈 㗉 㗊 㗋 㗌 㗍 㗎 㗏 㗐 㗑 㗒 㗓 㗔 㗕 㗖 㗗 㗘 㗙 㗚 㗛 㗜 㗝 㗞 㗟 㗠 㗡 㗢 㗣 㗤 㗥 㗦 㗧 㗨 㗩 㗪 㗫 㗬 㗭 㗮 㗯 㗰 㗱 㗲 㗳 㗴 㗵 㗶 㗷 㗸 㗹 㗺 㗻 㗼 㗽 㗾 㗿 㘀 㘁 㘂 㘃 㘄 㘅 㘆 㘇 㘈 㘉 㘊 㘋 㘌 㘍 㘎 㘏 㘐 㘑 㘒 㘓 㘔 㘕 㘖 㘗 㘘 㘙 㘚 㘛 㘜 㘝 㘞 㘟 㘠 㘡 㘢 㘣 㘤 㘥 㘦 㘧 㘨 㘩 㘪 㘫 㘬 㘭 㘮 㘯 㘰 㘱 㘲 㘳 㘴 㘵 㘶 㘷 㘸 㘹 㘺 㘻 㘼 㘽 㘾 㘿 㙀 㙁 㙂 㙃 㙄 㙅 㙆 㙇 㙈 㙉 㙊 㙋 㙌 㙍 㙎 㙏 㙐 㙑 㙒 㙓 㙔 㙕 㙖 㙗 㙘 㙙 㙚 㙛 㙜 㙝 㙞 㙟 㙠 㙡 㙢 㙣 㙤 㙥 㙦 㙧 㙨 㙩 㙪 㙫 㙬 㙭 㙮 㙯 㙰 㙱 㙲 㙳 㙴 㙵 㙶 㙷 㙸 㙹 㙺 㙻 㙼 㙽 㙾 㙿 㚀 㚁 㚂 㚃 㚄 㚅 㚆 㚇 㚈 㚉 㚊 㚋 㚌 㚍 㚎 㚏 㚐 㚑 㚒 㚓 㚔 㚕 㚖 㚗 㚘 㚙 㚚 㚛 㚜 㚝 㚞 㚟 㚠 㚡 㚢 㚣 㚤 㚥 㚦 㚧 㚨 㚩 㚪 㚫 㚬 㚭 㚮 㚯 㚰 㚱 㚲 㚳 㚴 㚵 㚶 㚷 㚸 㚹 㚺 㚻 㚼 㚽 㚾 㚿 㞀 㞁 㞂 㞃 㞄 㞅 㞆 㞇 㞈 㞉 㞊 㞋 㞌 㞍 㞎 㞏 㞐 㞑 㞒 㞓 㞔 㞕 㞖 㞗 㞘 㞙 㞚 㞛 㞜 㞝 㞞 㞟 㞠 㞡 㞢 㞣 㞤 㞥 㞦 㞧 㞨 㞩 㞪 㞫 㞬 㞭 㞮 㞯 㞰 㞱 㞲 㞳 㞴 㞵 㞶 㞷 㞸 㞹 㞺 㞻 㞼 㞽 㞾 㞿 㟀 㟁 㟂 㟃 㟄 㟅 㟆 㟇 㟈 㟉 㟊 㟋 㟌 㟍 㟎 㟏 㟐 㟑 㟒 㟓 㟔 㟕 㟖 㟗 㟘 㟙 㟚 㟛 㟜 㟝 㟞 㟟 㟠 㟡 㟢 㟣 㟤 㟥 㟦 㟧 㟨 㟩 㟪 㟫 㟬 㟭 㟮 㟯 㟰 㟱 㟲 㟳 㟴 㟵 㟶 㟷 㟸 㟹 㟺 㟻 㟼 㟽 㟾 㟿 㠀 㠁 㠂 㠃 㠄 㠅 㠆 㠇 㠈 㠉 㠊 㠋 㠌 㠍 㠎 㠏 㠐 㠑 㠒 㠓 㠔 㠕 㠖 㠗 㠘 㠙 㠚 㠛 㠜 㠝 㠞 㠟 㠠 㠡 㠢 㠣 㠤 㠥 㠦 㠧 㠨 㠩 㠪 㠫 㠬 㠭 㠮 㠯 㠰 㠱 㠲 㠳 㠴 㠵 㠶 㠷 㠸 㠹 㠺 㠻 㠼 㠽 㠾 㠿 㡀 㡁 㡂 㡃 㡄 㡅 㡆 㡇 㡈 㡉 㡊 㡋 㡌 㡍 㡎 㡏 㡐 㡑 㡒 㡓 㡔 㡕 㡖 㡗 㡘 㡙 㡚 㡛 㡜 㡝 㡞 㡟 㡠 㡡 㡢 㡣 㡤 㡥 㡦 㡧 㡨 㡩 㡪 㡫 㡬 㡭 㡮 㡯 㡰 㡱 㡲 㡳 㡴 㡵 㡶 㡷 㡸 㡹 㡺 㡻 㡼 㡽 㡾 㡿 㢀 㢁 㢂 㢃 㢄 㢅 㢆 㢇 㢈 㢉 㢊 㢋 㢌 㢍 㢎 㢏 㢐 㢑 㢒 㢓 㢔 㢕 㢖 㢗 㢘 㢙 㢚 㢛 㢜 㢝 㢞 㢟 㢠 㢡 㢢 㢣 㢤 㢥 㢦 㢧 㢨 㢩 㢪 㢫 㢬 㢭 㢮 㢯 㢰 㢱 㢲 㢳 㢴 㢵 㢶 㢷 㢸 㢹 㢺 㢻 㢼 㢽 㢾 㢿 㣀 㣁 㣂 㣃 㣄 㣅 㣆 㣇 㣈 㣉 㣊 㣋 㣌 㣍 㣎 㣏 㣐 㣑 㣒 㣓 㣔 㣕 㣖 㣗 㣘 㣙 㣚 㣛 㣜 㣝 㣞 㣟 㣠 㣡 㣢 㣣 㣤 㣥 㣦 㣧 㣨 㣩 㣪 㣫 㣬 㣭 㣮 㣯 㣰 㣱 㣲 㣳 㣴 㣵 㣶 㣷 㣸 㣹 㣺 㣻 㣼 㣽 㣾 㣿 㤀 㤁 㤂 㤃 㤄 㤅 㤆 㤇 㤈 㤉 㤊 㤋 㤌 㤍 㤎 㤏 㤐 㤑 㤒 㤓 㤔 㤕 㤖 㤗 㤘 㤙 㤚 㤛 㤜 㤝 㤞 㤟 㤠 㤡 㤢 㤣 㤤 㤥 㤦 㤧 㤨 㤩 㤪 㤫 㤬 㤭 㤮 㤯 㤰 㤱 㤲 㤳 㤴 㤵 㤶 㤷 㤸 㤹 㤺 㤻 㤼 㤽 㤾 㤿 㥀 㥁 㥂 㥃 㥄 㥅 㥆 㥇 㥈 㥉 㥊 㥋 㥌 㥍 㥎 㥏 㥐 㥑 㥒 㥓 㥔 㥕 㥖 㥗 㥘 㥙 㥚 㥛 㥜 㥝 㥞 㥟 㥠 㥡 㥢 㥣 㥤 㥥 㥦 㥧 㥨 㥩 㥪 㥫 㥬 㥭 㥮 㥯 㥰 㥱 㥲 㥳 㥴 㥵 㥶 㥷 㥸 㥹 㥺 㥻 㥼 㥽 㥾 㥿 㦀 㦁 㦂 㦃 㦄 㦅 㦆 㦇 㦈 㦉 㦊 㦋 㦌 㦍 㦎 㦏 㦐 㦑 㦒 㦓 㦔 㦕 㦖 㦗 㦘 㦙 㦚 㦛 㦜 㦝 㦞 㦟 㦠 㦡 㦢 㦣 㦤 㦥 㦦 㦧 㦨 㦩 㦪 㦫 㦬 㦭 㦮 㦯 㦰 㦱 㦲 㦳 㦴 㦵 㦶 㦷 㦸 㦹 㦺 㦻 㦼 㦽 㦾 㦿 㧀 㧁 㧂 㧃 㧄 㧅 㧆 㧇 㧈 㧉 㧊 㧋 㧌 㧍 㧎 㧏 㧐 㧑 㧒 㧓 㧔 㧕 㧖 㧗 㧘 㧙 㧚 㧛 㧜 㧝 㧞 㧟 㧠 㧡 㧢 㧣 㧤 㧥 㧦 㧧 㧨 㧩 㧪 㧫 㧬 㧭 㧮 㧯 㧰 㧱 㧲 㧳 㧴 㧵 㧶 㧷 㧸 㧹 㧺 㧻 㧼 㧽 㧾 㧿 㨀 㨁 㨂 㨃 㨄 㨅 㨆 㨇 㨈 㨉 㨊 㨋 㨌 㨍 㨎 㨏 㨐 㨑 㨒 㨓 㨔 㨕 㨖 㨗 㨘 㨙 㨚 㨛 㨜 㨝 㨞 㨟 㨠 㨡 㨢 㨣 㨤 㨥 㨦 㨧 㨨 㨩 㨪 㨫 㨬 㨭 㨮 㨯 㨰 㨱 㨲 㨳 㨴 㨵 㨶 㨷 㨸 㨹 㨺 㨻 㨼 㨽 㨾 㨿 㩀 㩁 㩂 㩃 㩄 㩅 㩆 㩇 㩈 㩉 㩊 㩋 㩌 㩍 㩎 㩏 㩐 㩑 㩒 㩓 㩔 㩕 㩖 㩗 㩘 㩙 㩚 㩛 㩜 㩝 㩞 㩟 㩠 㩡 㩢 㩣 㩤 㩥 㩦 㩧 㩨 㩩 㩪 㩫 㩬 㩭 㩮 㩯 㩰 㩱 㩲 㩳 㩴 㩵 㩶 㩷 㩸 㩹 㩺 㩻 㩼 㩽 㩾 㩿 㪀 㪁 㪂 㪃 㪄 㪅 㪆 㪇 㪈 㪉 㪊 㪋 㪌 㪍 㪎 㪏 㪐 㪑 㪒 㪓 㪔 㪕 㪖 㪗 㪘 㪙 㪚 㪛 㪜 㪝 㪞 㪟 㪠 㪡 㪢 㪣 㪤 㪥 㪦 㪧 㪨 㪩 㪪 㪫 㪬 㪭 㪮 㪯 㪰 㪱 㪲 㪳 㪴 㪵 㪶 㪷 㪸 㪹 㪺 㪻 㪼 㪽 㪾 㪿 㫀 㫁 㫂 㫃 㫄 㫅 㫆 㫇 㫈 㫉 㫊 㫋 㫌 㫍 㫎 㫏 㫐 㫑 㫒 㫓 㫔 㫕 㫖 㫗 㫘 㫙 㫚 㫛 㫜 㫝 㫞 㫟 㫠 㫡 㫢 㫣 㫤 㫥 㫦 㫧 㫨 㫩 㫪 㫫 㫬 㫭 㫮 㫯 㫰 㫱 㫲 㫳 㫴 㫵 㫶 㫷 㫸 㫹 㫺 㫻 㫼 㫽 㫾 㫿 㬀 㬁 㬂 㬃 㬄 㬅 㬆 㬇 㬈 㬉 㬊 㬋 㬌 㬍 㬎 㬏 㬐 㬑 㬒 㬓 㬔 㬕 㬖 㬗 㬘 㬙 㬚 㬛 㬜 㬝 㬞 㬟 㬠 㬡 㬢 㬣 㬤 㬥 㬦 㬧 㬨 㬩 㬪 㬫 㬬 㬭 㬮 㬯 㬰 㬱 㬲 㬳 㬴 㬵 㬶 㬷 㬸 㬹 㬺 㬻 㬼 㬽 㬾 㬿 㭀 㭁 㭂 㭃 㭄 㭅 㭆 㭇 㭈 㭉 㭊 㭋 㭌 㭍 㭎 㭏 㭐 㭑 㭒 㭓 㭔 㭕 㭖 㭗 㭘 㭙 㭚 㭛 㭜 㭝 㭞 㭟 㭠 㭡 㭢 㭣 㭤 㭥 㭦 㭧 㭨 㭩 㭪 㭫 㭬 㭭 㭮 㭯 㭰 㭱 㭲 㭳 㭴 㭵 㭶 㭷 㭸 㭹 㭺 㭻 㭼 㭽 㭾 㭿 㮀 㮁 㮂 㮃 㮄 㮅 㮆 㮇 㮈 㮉 㮊 㮋 㮌 㮍 㮎 㮏 㮐 㮑 㮒 㮓 㮔 㮕 㮖 㮗 㮘 㮙 㮚 㮛 㮜 㮝 㮞 㮟 㮠 㮡 㮢 㮣 㮤 㮥 㮦 㮧 㮨 㮩 㮪 㮫 㮬 㮭 㮮 㮯 㮰 㮱 㮲 㮳 㮴 㮵 㮶 㮷 㮸 㮹 㮺 㮻 㮼 㮽 㮾 㮿 㯀 㯁 㯂 㯃 㯄 㯅 㯆 㯇 㯈 㯉 㯊 㯋 㯌 㯍 㯎 㯏 㯐 㯑 㯒 㯓 㯔 㯕 㯖 㯗 㯘 㯙 㯚 㯛 㯜 㯝 㯞 㯟 㯠 㯡 㯢 㯣 㯤 㯥 㯦 㯧 㯨 㯩 㯪 㯫 㯬 㯭 㯮 㯯 㯰 㯱 㯲 㯳 㯴 㯵 㯶 㯷 㯸 㯹 㯺 㯻 㯼 㯽 㯾 㯿 㰀 㰁 㰂 㰃 㰄 㰅 㰆 㰇 㰈 㰉 㰊 㰋 㰌 㰍 㰎 㰏 㰐 㰑 㰒 㰓 㰔 㰕 㰖 㰗 㰘 㰙 㰚 㰛 㰜 㰝 㰞 㰟 㰠 㰡 㰢 㰣 㰤 㰥 㰦 㰧 㰨 㰩 㰪 㰫 㰬 㰭 㰮 㰯 㰰 㰱 㰲 㰳 㰴 㰵 㰶 㰷 㰸 㰹 㰺 㰻 㰼 㰽 㰾 㰿 㱀 㱁 㱂 㱃 㱄 㱅 㱆 㱇 㱈 㱉 㱊 㱋 㱌 㱍 㱎 㱏 㱐 㱑 㱒 㱓 㱔 㱕 㱖 㱗 㱘 㱙 㱚 㱛 㱜 㱝 㱞 㱟 㱠 㱡 㱢 㱣 㱤 㱥 㱦 㱧 㱨 㱩 㱪 㱫 㱬 㱭 㱮 㱯 㱰 㱱 㱲 㱳 㱴 㱵 㱶 㱷 㱸 㱹 㱺 㱻 㱼 㱽 㱾 㱿 㲀 㲁 㲂 㲃 㲄 㲅 㲆 㲇 㲈 㲉 㲊 㲋 㲌 㲍 㲎 㲏 㲐 㲑 㲒 㲓 㲔 㲕 㲖 㲗 㲘 㲙 㲚 㲛 㲜 㲝 㲞 㲟 㲠 㲡 㲢 㲣 㲤 㲥 㲦 㲧 㲨 㲩 㲪 㲫 㲬 㲭
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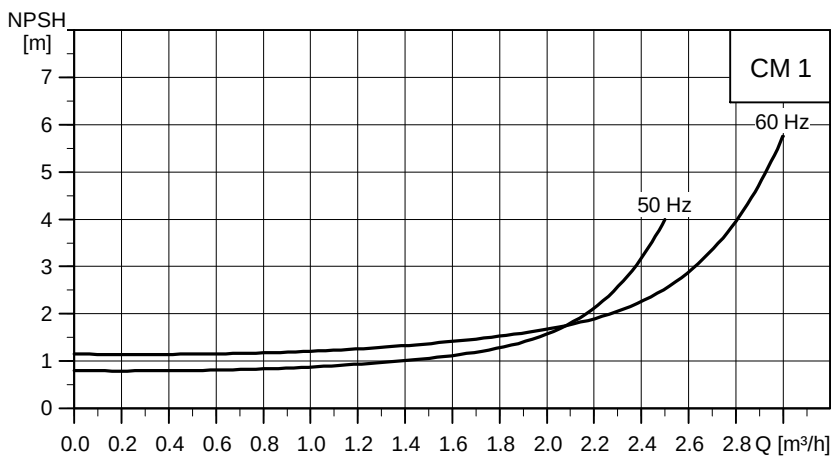


Fig. 4 NPSH curve for CM 1

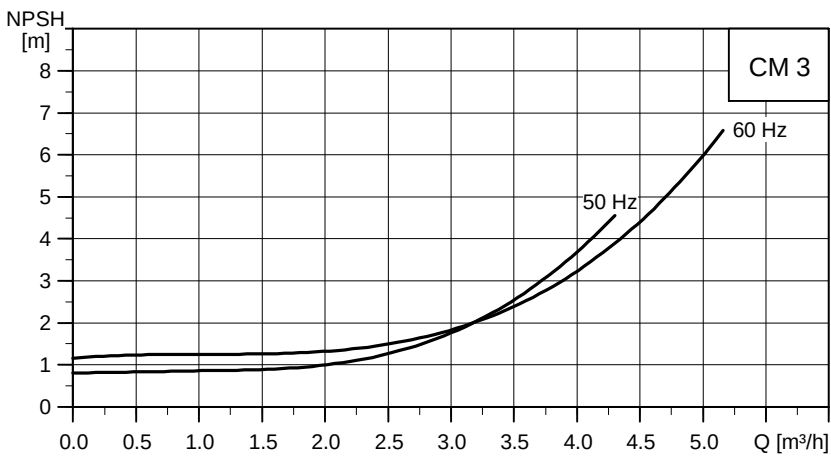


Fig. 5 NPSH curves for CM 3

TM04 0458 0309

TM04 0459 0309

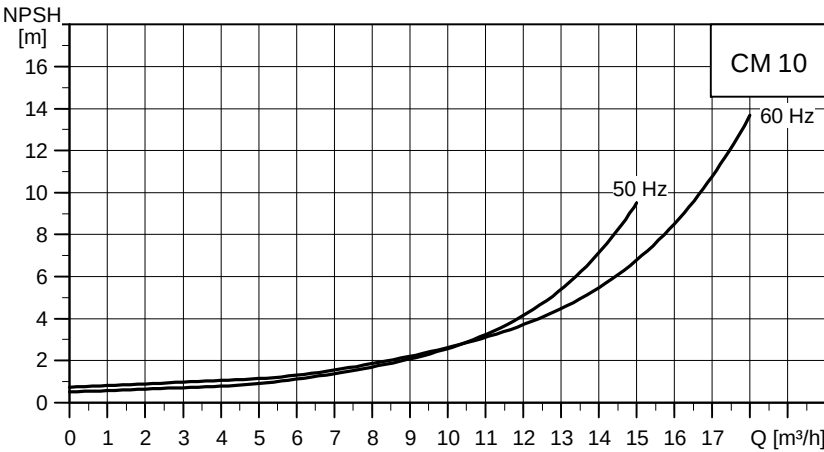
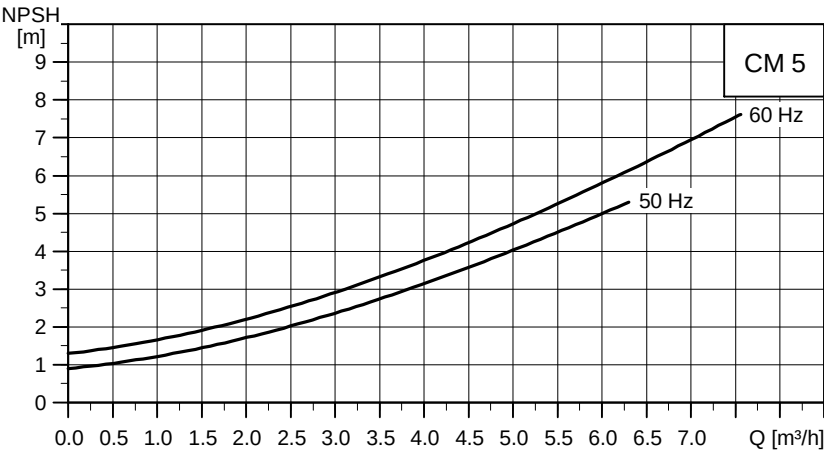


Fig. 6 NPSH curves for CM 10

TM04 0460 0309

TM04 0461 0309

TM04 0462 0309

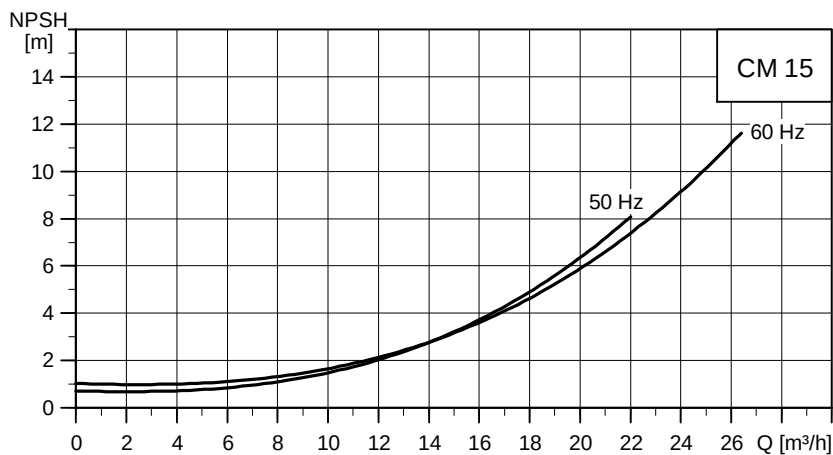


Fig. 7 NPSH curves for CM 15

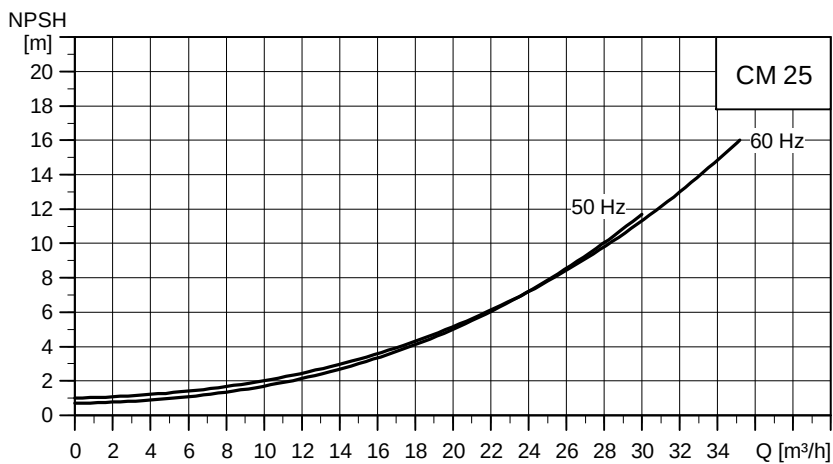


Fig. 8 NPSH curves for CM 25

TM04 0463 0309

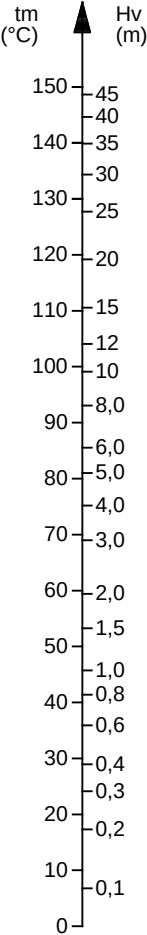


Fig. 9 Vapour pressure

TM00 3037 0800

Safety declaration

Please copy, fill in and sign this sheet and attach it to the pump returned for service.

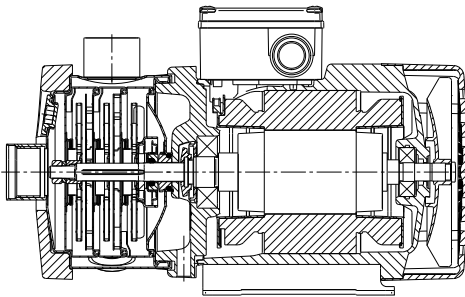
Media and application

Which media has the pump been used for: _____

In which application has the pump been used: _____

Fault description

If possible please make a circle around the faulty part.
(In case of an electrical fault, please mark the terminal box.)



Please give a short description of the fault:

We hereby declare that this product is free from hazardous chemicals, biological and radioactive substances.

Date and signature

Company stamp

TM04 0359 1008

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